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The Impact of Cold-Water Immersion on Post-Match Recovery in Trained Soccer Players: A Systematic Review and Meta-Analysis

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ABSTRACT

Cold-water immersion (CWI) is widely used by elite soccer players to enhance recovery after match play, yet systematic evidence supporting the effectiveness of this intervention is lacking. This systematic review with meta-analysis aimed to evaluate the effects of CWI on the recovery of physical performance, muscle damage, and delayed muscle soreness in trained soccer players after match play. A systematic database search was conducted and inclusion criteria were: (1) peer-reviewed, controlled trials; (2) competitive soccer players; (3) comparison of CWI with control/placebo after match/simulated match-play; (4) at least one of the following outcomes: 20 m sprint, countermovement jump (CMJ), leg strength via maximal voluntary contraction (MVC), creatine kinase (CK) for muscle damage, or delayed onset of muscle soreness (DOMS); and (5) outcome measurements taken 24, 48 or 72 h post-match. Random-effects meta-analyses were performed to calculate standardized mean differences (SMD) with 95% confidence intervals (CI) and prediction intervals (PI). Ten studies met the inclusion criteria. CWI significantly improved recovery of MVC (SMD = 1.02 [95% CI: 0.55 to 1.50; 95% PI: −0.48 to 2.53]) and CMJ (SMD = 0.38 [95% CI: 0.12 to 0.64; 95% PI: −0.39 to 1.15]), while sprint performance remained unaffected (SMD = −0.59 [95% CI: −1.41 to 0.23; 95% PI: −4.14 to 2.95]). CWI reduced CK levels (SMD = −0.77 [95% CI: −1.17 to −0.37; 95% PI: −2.23 to 0.69]) and alleviated DOMS (SMD = −1.04 [95% CI: −1.91 to −0.17; 95% PI: −4.12 to 2.03]). MVC and CK recovery improved consistently across all time points, while DOMS relief was observed at 24 and 72 h post-match. CMJ benefits were only evident at 48 h post-match. However, as the 95% PIs for all outcomes included the null effect, the observed benefits should be interpreted cautiously, as future studies may yield smaller or null effects. Overall, CWI may enhance recovery of muscle strength, reduce muscle damage, and alleviate soreness in trained male and female soccer players, but does not seem to impact sprint performance, and its effect on CMJ is time-dependent.

1 | Introduction

Soccer is a physically demanding sport where participation in match play induces acute fatigue, characterized by reduced physical performance, increased muscle damage and delayed muscle soreness (DOMS). Depending on the match load, it may take several days for players to fully recover [1–5]. Throughout the competitive season, elite players typically engage in at least one match every week. However, during international and domestic cup league games, matches may be scheduled bi-weekly

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or even tri-weekly. Moreover, the prevalence of extra time (120 min) matches has increased due to a rise in the number of knock-out games in international tournaments [6]. During these congested periods, the allocated time for rest and recovery between matches might be compromised, potentially increasing the accumulation of fatigue, underperformance and injury [7]. Therefore, optimizing the recovery processes becomes crucial [4, 8].

In addition to rest, nutrition, and sleep, cold-water immersion (CWI) has emerged as one of the most popular post-match recovery interventions [9]. Post-exercise CWI typically involves partial or full body immersion in cold water ranging from 5°C to 15°C for a period of 5 to 20 min [8]. This practice is commonly administered immediately after exercise or match termination, either as a single bout or with multiple bouts separated by short rest periods [10–12]. The widespread popularity of CWI among soccer players stems from the belief that CWI alleviates muscle soreness and perception of fatigue and thereby accelerates the recovery of physical performance and consequently a return to play in a better condition. This potential recovery benefit is thought to arise from CWI's ability to reduce exercise-induced inflammation and muscle damage [10]. Despite the widespread popularity of CWI and the theoretical benefits associated with its use, its effectiveness on post-exercise recovery remains inconclusive [13].

Although the effects of CWI on recovery after exercise have been the scope of several previous reviews and meta-analyses [14–17], it is well known that the mechanisms of exercise-induced fatigue and recovery are task-dependent [18–20]. Therefore, it is surprising that, to date, no meta-analysis has specifically investigated the effects of CWI on recovery in soccer, the world's largest sport. Notably, focusing on team sports in general, Higgins and colleagues [21] showed that CWI leads to favorable post-exercise outcomes compared to control in both the countermovement jump and sprint performance 24 h post-exercise, and in the perception of fatigue 72 h post-exercise, with no significant beneficial effects on other time points. Furthermore, there was no effect on DOMS or muscle damage measured by creatine kinase levels. As this study included soccer among other team sports like volleyball and basketball in a pooled analysis, sports whose neuro-physiological load and muscle involvement differ substantially from soccer [22], no certain conclusions can be made regarding the effect of post-exercise CWI on recovery in soccer athletes. Indeed, comparative studies have shown prolonged impairments in sprint performance and elevated CK levels in soccer and rugby players compared to other team sports [20, 23], supporting a sport-specific fatigue–recovery profile. Moreover, soccer has among the highest injury rates in team sports [24], and CWI is already widely adopted in professional soccer, with reports indicating that up to 90% of elite teams are implementing it as a core recovery strategy [25]. Together, these factors justify a focused meta-analysis on CWI recovery efficacy specifically within soccer.

Therefore, our systematic review and meta-analyses aim to address this gap by specifically evaluating the effects of CWI on physical performance, muscle damage, and DOMS in trained soccer players following match play.

2 | Materials and Methods

2.1 | Study Design

This systematic review with meta-analyses followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [26], and was prospectively registered at PROSPERO [CRD42024543581] on 26 May 2024.

2.2 | Search Strategy and Selection Criteria

A systematic literature search using the PICO framework was conducted by JV from January 15 to 19, 2024, and again in January 2025, in the PubMed, Cochrane, Scopus, and SPORTDiscus databases. A combination of keywords (supplement 1) was used for the following concepts: cold water immersion, soccer, football, and recovery. The initial search across these databases yielded 260 studies. After removing duplicates, 127 articles remained, and their abstracts were screened for eligibility by JV and subsequently verified by CB. Following abstract screening, 55 full-text articles were retrieved and independently assessed for eligibility by two reviewers (JV and CB), resulting in the inclusion of 8 studies that met the predefined criteria. Additionally, through review of other systematic reviews and meta-analyses on the topic, 2 additional studies were identified and included. Any discrepancies regarding study inclusion were resolved by consulting a third author (PE) to reach a consensus. In total, 10 randomized controlled trials (RCTs) were selected for inclusion in this meta-analysis (Figure 1).

Inclusion criteria were defined as follows: (1) Studies had to be peer-reviewed randomized controlled trials available in full text and published in English; (2) Participants were competitive male or female soccer players aged 15 to 45 years; (3) CWI administered as a post-exercise recovery strategy for 5–20 min at 5°C–15°C following a soccer match, simulated match or activity mimicking the intensity/volume of a soccer match and including at least a CWI group; (4) Studies utilized a passive control or placebo control group for comparison with the CWI group. Outcome measures included at least one of the following physical performance indicators: 20 m sprint, countermovement jump (CMJ), or maximal leg muscle strength assessed by maximal voluntary contraction (MVC); muscle damage measured by creatine kinase (CK); or pain perception assessed as delayed onset of muscle soreness (DOMS); (5) Assessments performed before match/simulated match play and at least at one of the following time points, 0 h, 24 h (from 21 to 25 h), 48 h (from 45 to 49 h) and 72 h (from 69 to 73 h) post-exercise. Studies with combined or different treatments were excluded from the analyses. An overview of the search process and reasons for exclusion can be found in (Figure 1).

2.3 | Data Extraction

Data regarding publication details, methodology, participant characteristics, treatment and control protocol, and assessment measures at various time points were extracted and recorded in Excel by one author (JV) and double-checked by a second author (CB). In cases where information was incomplete, attempts were

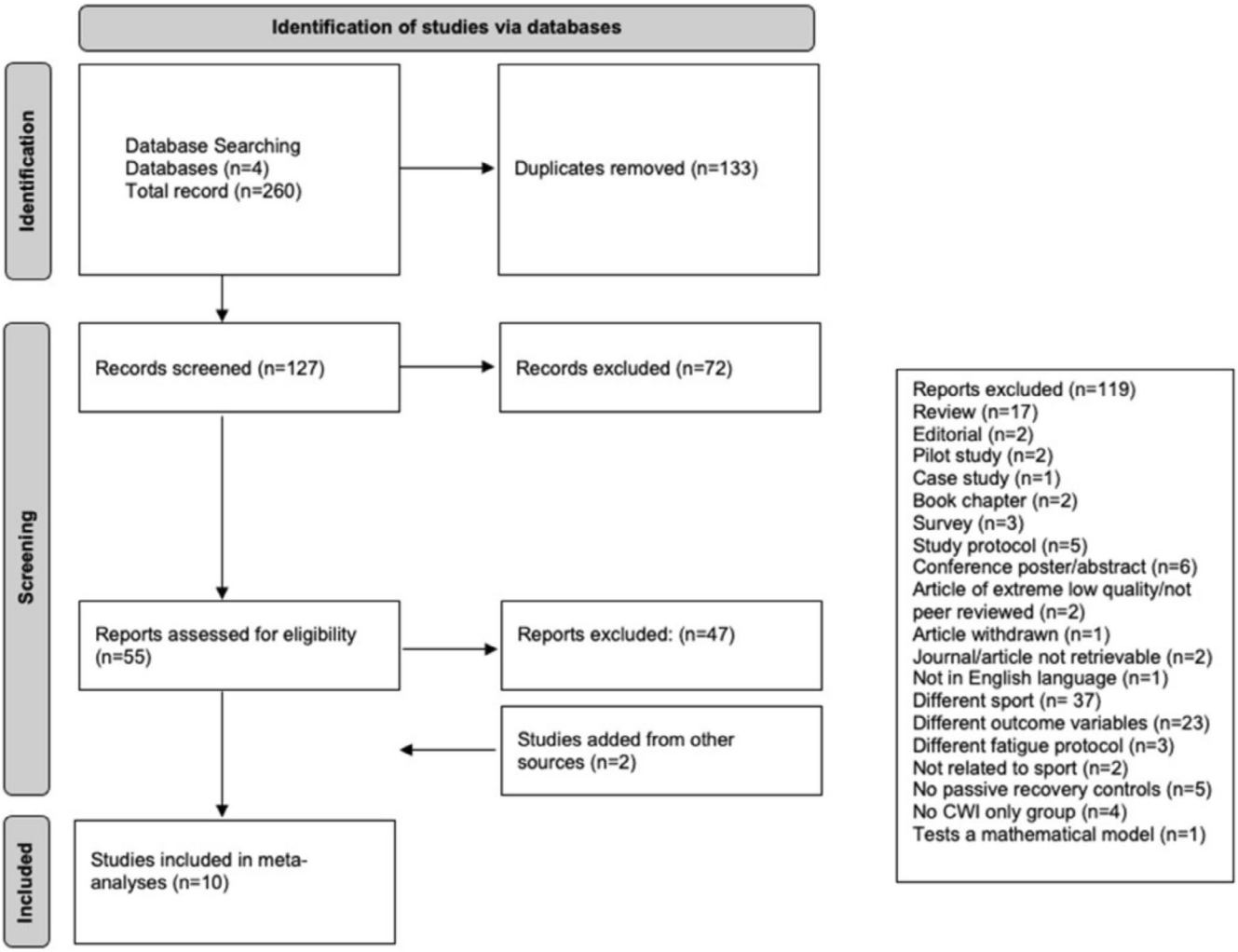


FIGURE 1 | Flow chart showing the different phases of literature search according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta Analyses). CWI, Cold-water immersion.

made to contact the original authors via e-mail to obtain missing data. When direct access to the data was not feasible, information was manually extracted from the figures using Plotdigitizer (plotdigitizer.com).

2.4 | Quality Assessment and Risk of Bias

The methodological quality and bias of the included studies were assessed using the Physiotherapy Evidence Database (PEDro) Scale [27]. This scale evaluates 11 items (first question not scored): Inclusion criteria and source, random allocation, concealed allocation, baseline similarity, subject blinding, therapist blinding, assessor blinding, completeness of follow-up, intention-to-treat analysis, between-group statistical comparisons, *and* point measures and variability. Studies were categorized based on their methodological quality as follows: 0–3 poor quality, 4–5 fair quality, 6–8 good quality, and 9–10 excellent quality [28]. Studies were included if they scored a minimum of 5 out of 10, indicating a low risk of bias [29]. The assessment of methodological quality and bias was conducted independently by two authors (JV and CB). In cases of disagreement, a third author (PE) was consulted to reach a consensus.

2.5 | Statistical Methods

Studies included in the meta-analysis were restricted to those with fewer than three domains of bias on the Pedro scale, which assesses methodological quality based on factors such as randomization, blinding, and allocation concealment. Studies exceeding this threshold were excluded due to high or unclear risk of bias, primarily stemming from conflicts of interest and lack of blinding of investigators and/or participants (see online Supporting Information for details). Publication bias was assessed qualitatively rather than using a funnel plot or formal statistical tests, as fewer than ten studies were available for each time point. Heterogeneity was evaluated using the I^2 statistic, with thresholds defined as low (0%–25%), moderate (26%–50%), high (51%–75%), and very high (> 75%) heterogeneity [30]. The presence of significant heterogeneity was further evaluated using the Q-test, with a $p < 0.10$ for the Q test considered indicative of significant heterogeneity [31].

Effect sizes were calculated as standardized mean differences (SMDs) with 95% confidence intervals (CIs) and 95% prediction intervals (PIs) to compare the effects of CWI and control recovery. For some subgroup analyses based on fewer than three studies, PI was not calculated. A random-effects model was employed to

TABLE 1 | Overview of included studies.

Study	Study type	Subject N, sex	Fatigue protocol	CWI group	Outcome measures included in the meta analyses	Timing of measures
Rowsell et al. 2009	Between groups design	20 males	Soccer match	10°C; 5 × 1 min	20 m sprint, CMJ, CK	24 h
Bouzid et al. 2018	Cross-over design	8 males	Modification of the original Loughborough Intermittent Shuttle Test	10°C; 10 min	20 m sprint, CMJ, MVC, CK, DOMS	24 h, 48 h, 72 h
Nasser et al. 2023	Cross-over design	12 males	Loughborough Intermittent Shuttle Test	11.3°C; 15 min	20 m sprint, CMJ, CK, DOMS	24 h, 48 h
Ascensao et al. 2011	Between groups design	20 males	Soccer match	10°C; 10 min	20 m sprint, CMJ, MVC, CK, DOMS	24 h, 48 h
Rupp et al. 2012	Between groups design	13 males, 9 females	YoYo intermittent recovery test	12°C; 15 min	CMJ	24, 48 h
Bouchiba et al. 2021	Between groups design	12 males	Simulated soccer match	10°C; 10 min	20 m sprint, CMJ, MVC, CK	24, 48, 72 h
Coelho et al. 2021	Between groups design	25 males	Soccer match	10°C; 10 min	20 m sprint, CMJ, CK, DOMS	24, 48 h
Farkhani et al. 2016	Between groups design	30 males	Simulated soccer match	10°C–15°C; 10 min	20 m sprint	24 h
Roonkiani et al. 2020	Between groups design	18 males	Simulated soccer match	8°C; 10 min	CK	24, 48 h
Gustafsson et al. 2025	Between groups design	43 males	Simulated soccer match	10°C; 10 min	20 m sprint, CMJ, MVC	24, 48 h

Abbreviations: 20 m sprint, 20 m sprint; CK, creatine kinase; CMJ, countermovement jump; DOMS, delayed onset of muscle soreness; MVC, maximal voluntary contraction.

pool the effects when heterogeneity was present, with individual study effects weighted using the inverse variance method. The DerSimonian and Laird method was used to estimate between-study variance. SMDs were classified as trivial (SMD < 0.20), small (SMD 0.20–0.60), moderate (SMD 0.61–1.20), large (SMD 1.21–2.00), and very large (SMD > 2.01) [32] based on established guidelines. The robustness of the overall effect estimates was examined by comparing results from fixed-effect and random-effects models. Separate analyses were conducted for performance variables and muscle damage indicators, including MVC, CMJ, sprint performance, DOMS, and CK levels. Subgroup analyses were performed for different time points (24 h, 48 h, 72 h) to determine whether effects varied across assessed variables and recovery times. The main results are presented in tables and visualized using forest plots, showing individual study effect sizes and the pooled effect estimate, with relevant statistics provided alongside. All statistical analyses were performed using Stata version 18.5 (StataCorp, College Station, Texas, USA).

3 | Results

3.1 | Overview of Included Studies

Following a systematic search across four electronic databases, a total of 10 studies published between 2009 and 2025 were identified and included in the meta-analysis. All studies employed either a crossover or a between-group design. A summary of the key characteristics of the included studies is presented in (Table 1).

3.2 | Risk of Bias and Article Quality

PEDro scores of the included articles are presented in (Table S1). Based on the PEDro scale, two studies [33, 34], were classified as fair quality, while eight were rated as good quality [35–42]. In none of the included studies were participants and/or therapists blinded to treatment (questions 5 and 6). All included studies had a score of 5 or higher, indicating a low risk for bias [29].

3.3 | Meta-Analyses

3.3.1 | Effect of CWI on MVC

(Figure 2), shows the effect of CWI on the recovery of MVC following a soccer match. Overall, CWI significantly improved recovery of MVC compared to passive rest (i.e., control) ($p = 0.001$) (SMD = 1.02 [95% CI: 0.55 to 1.50; 95% PI: −0.48 to 2.53]). Further analysis revealed a significant positive effect of CWI on MVC recovery at all assessed time points: 24 h ($p = 0.001$) (SMD = 1.22 [95% CI: 0.48 to 1.97; 95% PI: −1.83 to 4.28]), 48 h ($p = 0.05$) (SMD = 1.10 [95% CI: −0.02 to 2.21; 95%

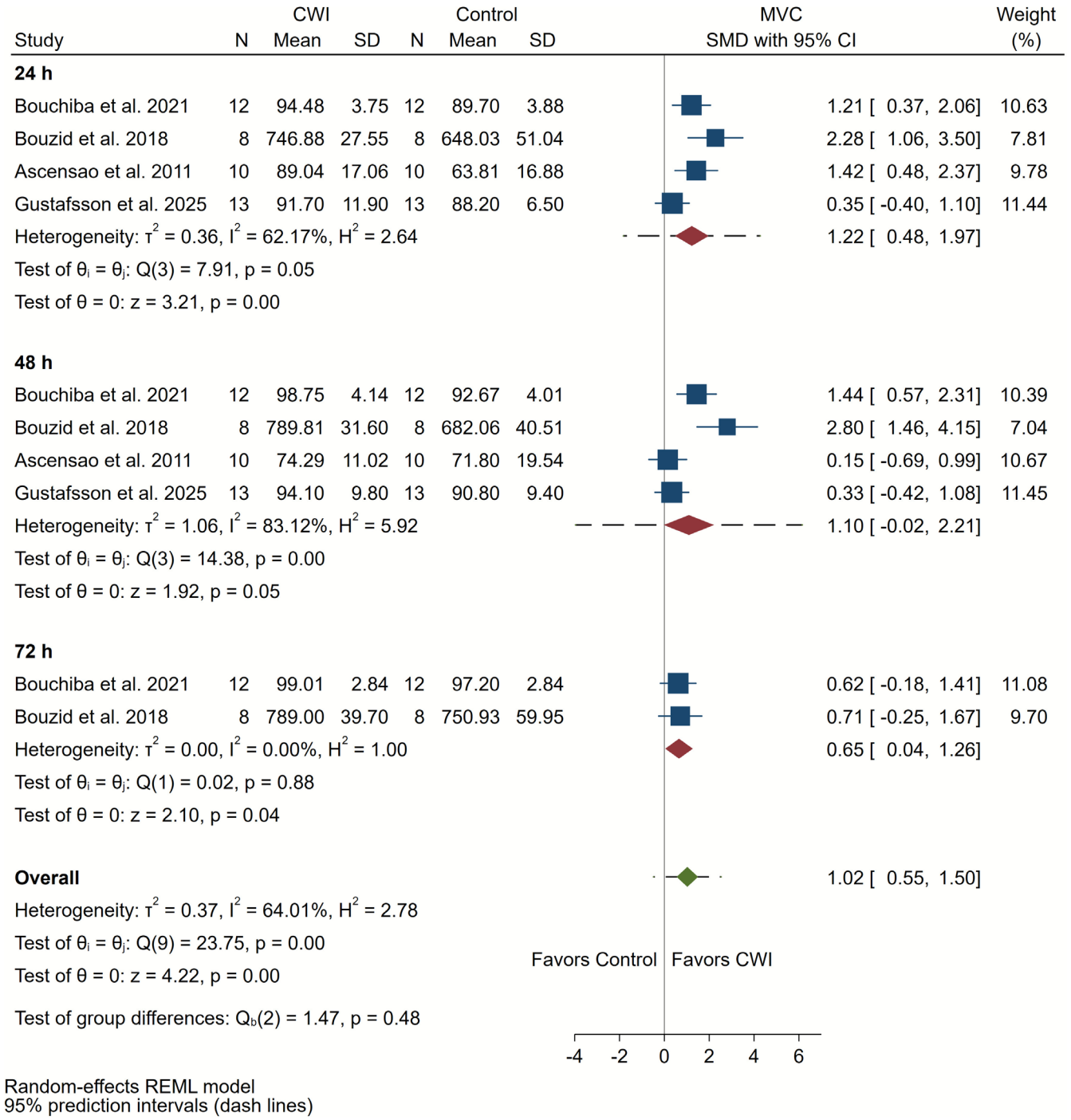


FIGURE 2 | Forest plots showing the effect of CWI on MVC following soccer exercise at various time points. CWI: Cold-water immersion, MVC: Maximal voluntary contraction, CI: Confidence interval, SMD: Standardized mean difference.

PI: −2.96 to 6.15]), and 72 h ($p = 0.04$) (SMD = 0.65 [95% CI: 0.04 to 1.26]).

3.3.2 | Effect of CWI on CMJ

(Figure 3), shows the effect of CWI on CMJ performance after the soccer match. Overall, CWI significantly improved recovery of CMJ compared to passive rest (i.e., control) ($p < 0.01$) (SMD = 0.38 [95% CI: 0.12 to 0.64; 95% PI: −0.39 to 1.15]). However, further analysis revealed that CWI was only beneficial for CMJ recovery at 48 h ($p = 0.01$) (SMD = 0.53 [95% CI: 0.10 to 0.96; 95% PI: −0.59 to 1.66]). At 24 and 72 h, CWI did not reach a significant difference compared to passive rest $p = 0.15$ (SMD = 0.22 [95% CI: −0.08 to −0.51; 95% PI: −0.15 to 0.58]) and $p = 0.53$ (SMD = 0.49 [95% CI: −1.03 to 2.02]) respectively.

3.3.3 | Effect of CWI on 20-m Sprint

(Figure 4), shows the effect of CWI on 20 m sprint performance after soccer exercise. Overall, CWI did not significantly improve

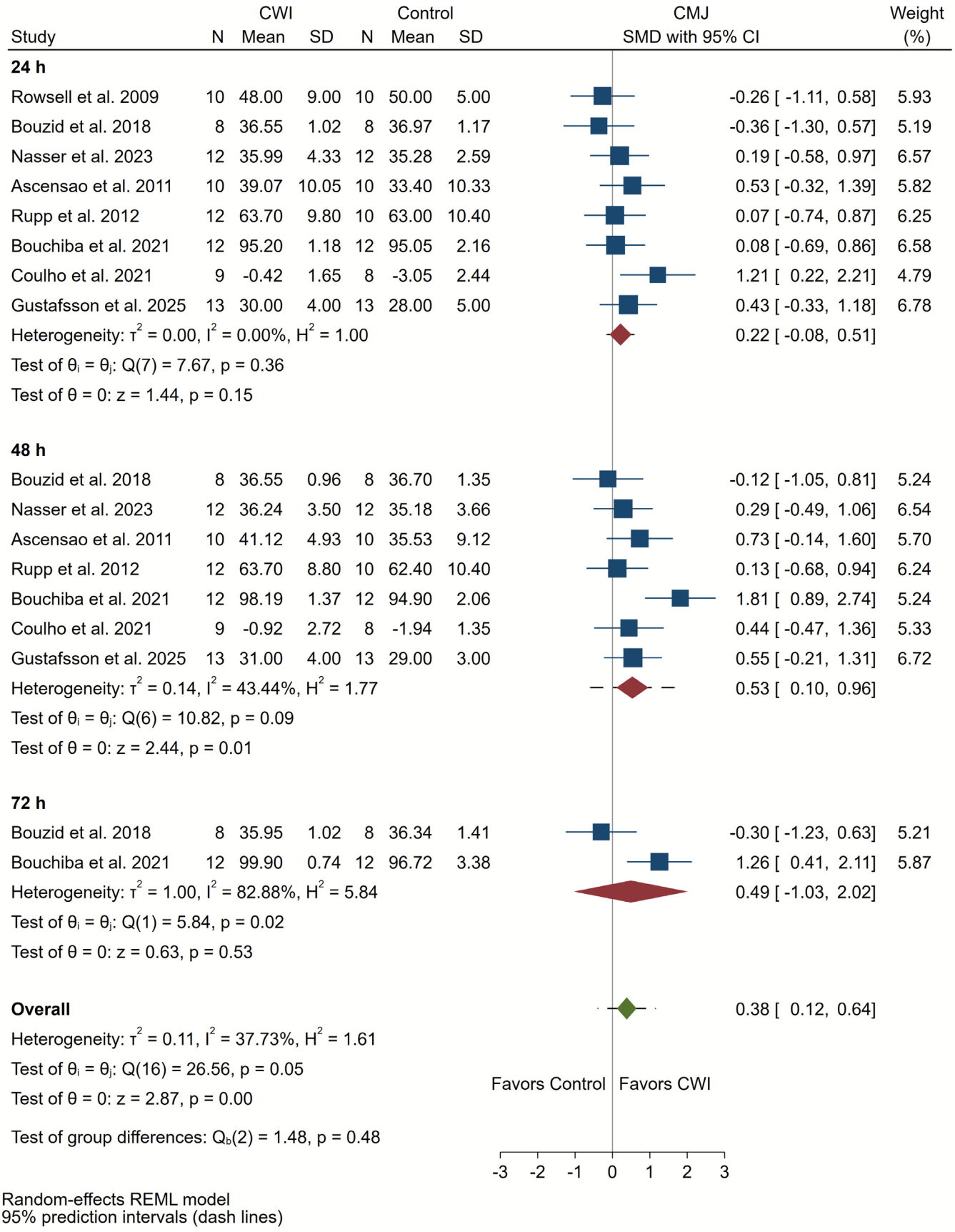


FIGURE 3 | Forest plots showing the effect of CWI on CMJ following soccer exercise at various time points. CWI: Cold-water immersion, CMJ: Countermovement jump, CI: Confidence interval, SMD: Standardized mean difference.

recovery of 20 m sprint performance compared to passive rest ($p = 0.16$) (SMD = −0.59 [95% CI: −1.41 to 0.23; 95% PI: −4.14 to 2.95]). Similarly, no beneficial effect of CWI on 20-m sprint recovery was found at any of the assessed time points (24 h, $p = 0.51$) (SMD = −0.44 [95% CI: −1.75 to 0.87; 95% PI: −5.13 to 4.25]); 48 h, $p = 0.24$ (SMD = −0.67 [95% CI: −1.78 to 0.44; 95% PI: −4.60 to 3.26]), and 72 h, $p = 0.69$ (SMD = −0.86 [95% CI: −5.14 to 3.42]) respectively.

3.3.4 | Effect of CWI on Creatine Kinase

(Figure 5) shows the effect of CWI on CK levels after soccer match-play. Overall, CWI reduced post-exercise levels of CK levels more compared to passive rest ($p < 0.01$) (SMD = −0.77 [95% CI: −1.17 to −0.37; 95% PI: −2.23 to 0.69]). Further analysis revealed that this reduction in CK following CWI compared to passive rest was evident at all assessed time points (24 h, $p = 0.04$) (SMD = −0.86 [95% CI: −1.68 to −0.04; 95% PI: −3.66 to 1.94]); 48 h, $p = 0.01$ (SMD = −0.69 [95% CI: −1.21 to −0.17; 95% PI: −2.22 to 0.84]), and 72 h, $p = 0.01$ (SMD = −0.82 [95% CI: −1.44 to −0.20]) respectively.

3.3.5 | Effect of CWI on DOMS

(Figure 6) shows the effect of CWI on DOMS after soccer exercise. Overall, CWI led to a significantly lower DOMS compared to passive rest ($p = 0.02$) (SMD = −1.04 [95% CI: −1.91 to −0.17; 95% PI: −4.12 to 2.03]). Further analysis revealed that this beneficial effect on DOMS was evident at both 24 h ($p = 0.01$) (SMD = −1.71 [95% CI: −2.98 to −0.43; 95% PI: −7.50 to 4.09]) and 72 h ($p = 0.04$) (SMD = −1.02 [95% CI: −2.01 to −0.03]), while no differences compared to passive rest were seen at 48 h post-exercise ($p = 0.56$) (SMD = −0.41 [95% CI: −1.77 to 0.95; 95% PI: −6.74 to 5, 95]).

Although the pooled effects for several outcomes were statistically significant based on the 95% CIs, the corresponding 95% PIs all included the null value. This suggests that the addition of future studies could plausibly shift the overall estimates toward smaller or even null effects, indicating that the observed benefits of CWI may not be consistently reproducible across different study settings.

4 | Discussion

This systematic review with meta-analysis is, to our knowledge, the first to specifically evaluate the effects of CWI on post-match recovery in soccer players, making it highly relevant to teams considering or currently using CWI as part of their recovery protocols. Our primary finding indicates that post-match CWI may enhance recovery of physical performance (MVC and jump performance), alleviate perceived pain and fatigue (DOMS), and lower biochemical markers of muscle damage (CK) in the days following match play. Overall, this review supports CWI as a potentially effective strategy to accelerate recovery of physical performance, reduce muscle damage, and alleviate pain in soccer players.

Soccer presents a distinct physiological profile, combining high volumes of eccentric muscle loading from repeated sprinting, decelerations, changes of direction, and jumping, sustained over 90 min with limited substitutions and large playing areas. This leads to substantial neuromuscular and metabolic fatigue, differentiating soccer from other team sports with shorter durations, smaller playing areas, and more frequent substitutions [43]. In turn, accumulated fatigue and muscle damage may require several days to fully recover from and are associated with increased injury risk, highlighting the importance of effective recovery strategies. Our analysis examined various markers of physical performance (MVC, CMJ, and sprint performance) to assess the impact of CWI on physical recovery. Results showed that CWI effectively facilitated MVC recovery at 24-, 48-, and 72 h post-exercise, and improved CMJ recovery at 48 h post-exercise, though it did not significantly affect 20 m sprint performance. This discrepancy may be attributed to differences in measurement sensitivity, as MVC and CMJ tests often have lower variability than sprint performance [44]. Another possibility is that CWI more effectively supports recovery of isolated muscle contractions (such as during MVC and CMJ) than from repetitive contraction cycles like those in sprinting. This idea aligns with Choo and colleagues [45] who also observed improved recovery in muscle strength but not in sprint performance following high-intensity exercise in general. Our finding that CWI accelerates recovery of physical performance compared to passive rest aligns partly with an earlier review by Higgins and colleagues [21], which reported positive effects of CWI on CMJ and sprint performance in team sports, though only at 24 h post-exercise. However, Higgins and colleagues [21] included a wide range of team sports beyond soccer, such as American Football, basketball and volleyball. As these sports differ, to a smaller or larger extent, from soccer in their physical and metabolic demands, this diversity likely influenced the outcomes of their pooled analysis. In contrast, our soccer-specific findings offer more targeted and sport-relevant insights for players and coaches.

CK is a well-established marker for muscle damage and is known to increase significantly following physically demanding activities, including soccer match play. Although sometimes debated [46], CK is, in general, considered an important indicator of muscle damage and fatigue, and therefore, interesting to monitor during recovery [47]. In our meta-analyses, we found that post-match CWI can accelerate CK recovery kinetics when compared to passive rest at all assessed time points. This contrasts with the findings by Higgins and colleagues [21], who reported no significant effect of CWI on CK in team sports overall. However, as previously discussed, their meta-analysis encompassed a broad spectrum of team sports with varying physical demands, not addressing sport-specific outcomes. Moreover, the inclusion of recent studies since Higgins' review has advanced our understanding, with most studies included in our analysis dating from 2017 and onwards. Indeed, a recent review by Moore and colleagues [48], on the effect of CWI on CK after high-intensity exercise in general supports our finding that CWI accelerated the recovery of exercise-induced increases in CK. This may, in turn, suggest that the observed effect of post-exercise CWI on neuromuscular performance and DOMS may be due to improved time-course recovery at the biochemical level.

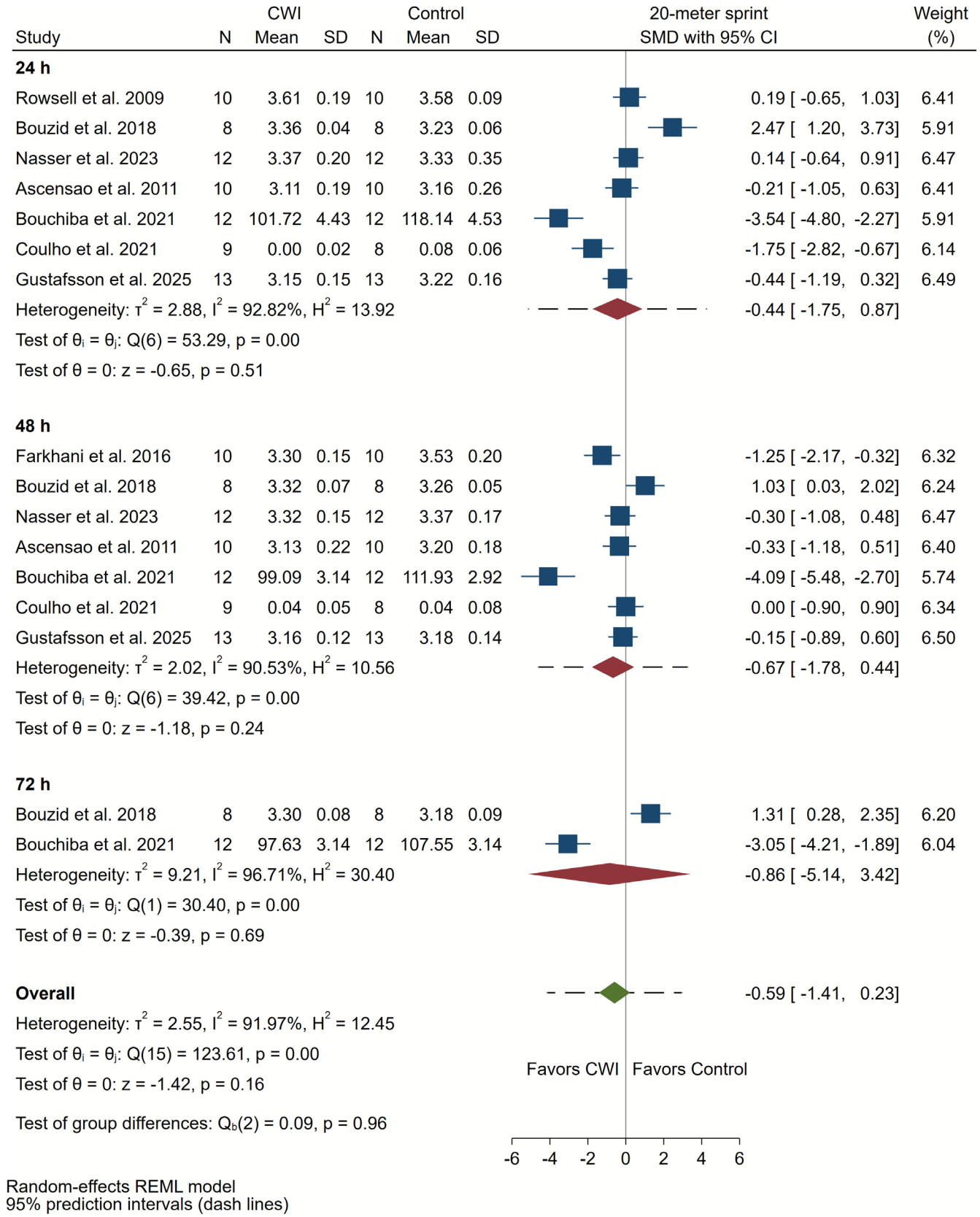


FIGURE 4 | Forest plots showing the effect of CWI on 20 m sprint performance following soccer exercise at various time points. CWI: Cold-water immersion, 20 m sprint: 20 m sprint, CI: Confidence interval, SMD: Standardized mean difference.

Assessment of DOMS through subjective rating gives important insight into players' perception of fatigue and soreness, and as such, feelings may impair performance. Monitoring post-exercise DOMS gives imperative guidance to determine future training loads [49]. Our meta-analysis indicates that post-match CWI provides an analgesic effect, as a subjective rating of

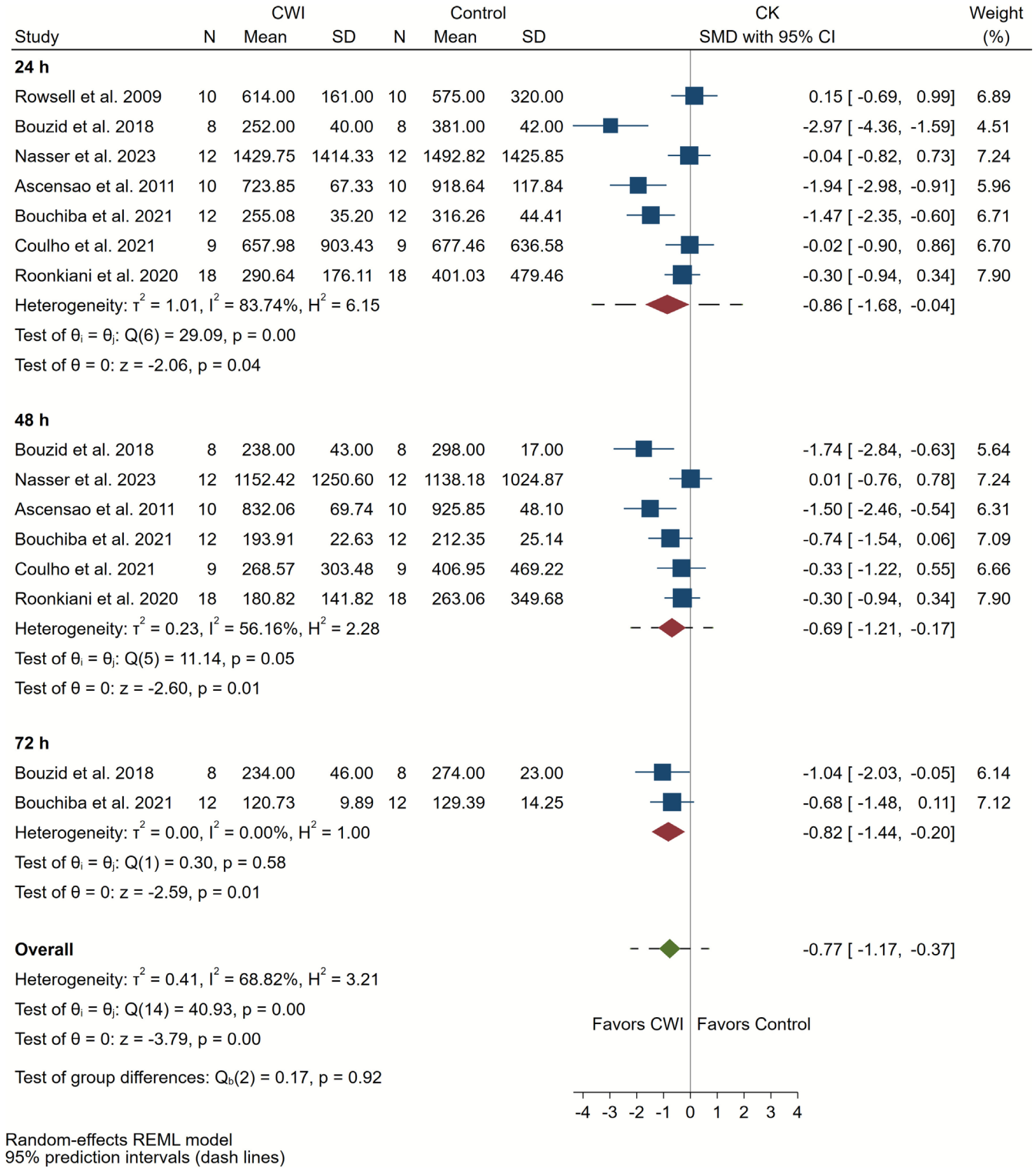


FIGURE 5 | Forest plots showing the effect of CWI on CK levels following soccer exercise at various time points. CWI: Cold-water immersion, CK: Creatine kinase, CI: Confidence interval, SMD: Standardized mean difference.

DOMS was lower among players after CWI compared to passive rest, at least if assessed 24 or 72 h after match. Nevertheless, it is important to acknowledge that the output for the 72 h time point is based on a single observation, and therefore, caution should be taken when drawing conclusions. Our findings are in contrast with Higgins and colleagues [21], who did not find any positive effect of CWI on DOMS in team sports in general, but in coherence with more recent reviews on high-intensity exercise in general [45, 48].

It is important to emphasize that the effectiveness of CWI was more pronounced for outcome variables that exhibited substantial fatigue or muscle damage following the simulated soccer match, such as MVC, CK, and DOMS. This trend was

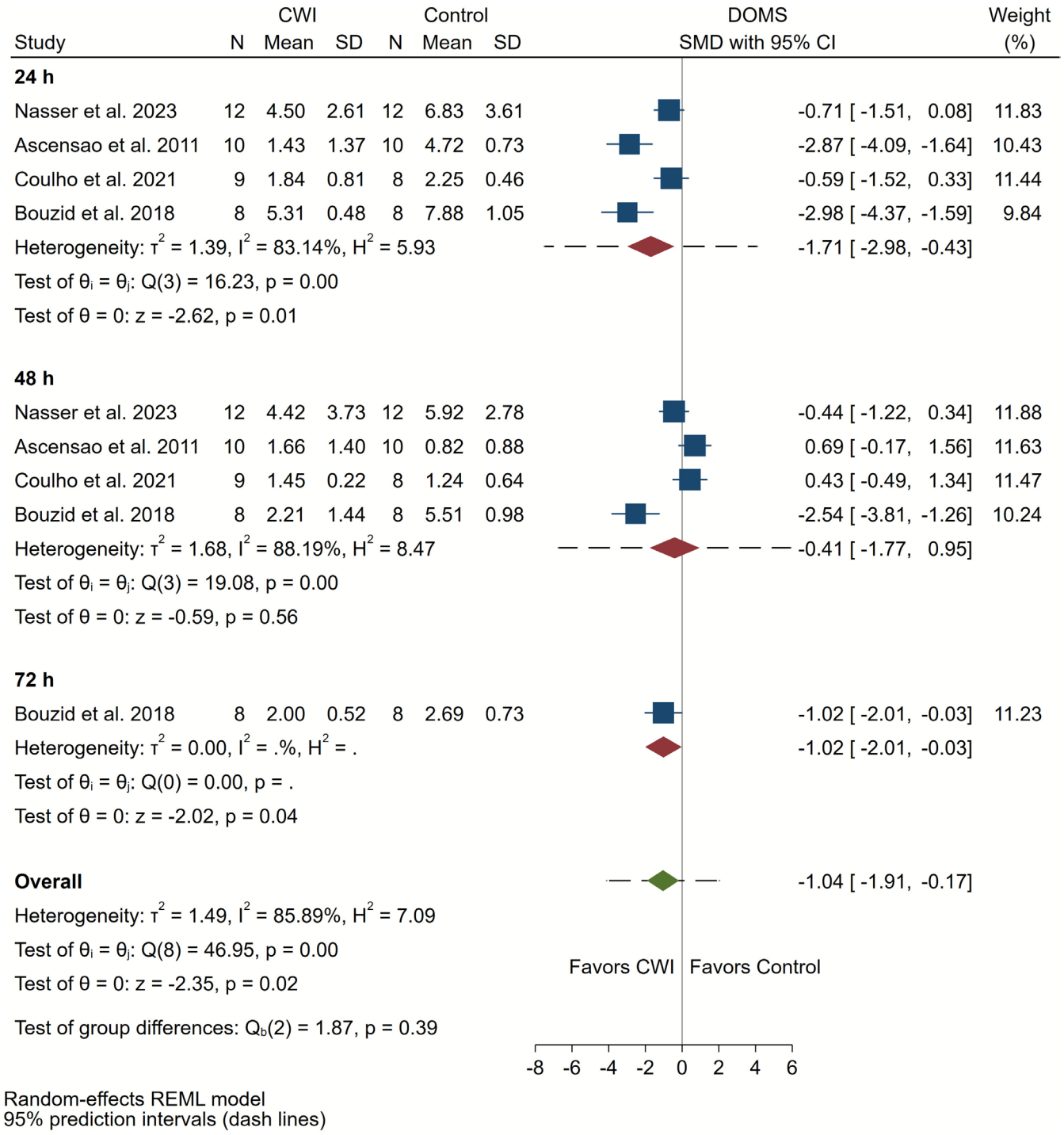


FIGURE 6 | Forest plots showing the effect of CWI on DOMS following soccer exercise at various time points. CWI: Cold-water immersion, DOMS: Delayed onset of muscle soreness, CI: Confidence interval, SMD: Standardized mean difference.

clearly reflected in the meta-analyses, where CWI showed greater benefits at time points when these markers were most affected in the control groups. Notably, 20 m sprint performance demonstrated only minor impairments under control conditions, which corresponded with a small and statistically non-significant effect of CWI on this measure. These findings suggest that the magnitude of CWI's efficacy is linked to the extent of physiological disturbance induced by the preceding exercise bout.

This systematic review with meta-analyses provides detailed insights into the effects of post-match CWI on recovery among trained soccer players following exhaustive, soccer-specific activities such as matches and simulated play. By analyzing key performance outcomes, including neuromuscular performance, biochemical markers of muscle damage, and perceived DOMS ratings, it provides novel, valuable, and reliable information for soccer players and practitioners. Nonetheless, this meta-analysis has limitations. Firstly, the nature of CWI treatments makes

true blinding of participants and researchers impossible, potentially introducing bias. Future studies should consider including placebo groups with sham therapies to mitigate this issue. Additionally, the relatively small sample sizes in the included studies limit statistical power.

Another consideration is the inclusion of studies that use both actual soccer match play and simulated soccer protocols. While this may introduce some variability in physiological responses, several of the simulated protocols, such as the Loughborough Intermittent Shuttle Test and the Copenhagen Soccer Test, have previously been validated and shown to elicit neuromuscular and biochemical responses comparable to those observed in competitive matches [50, 51]. Although the Yo-Yo Intermittent Recovery Test is shorter in duration and may induce different fatigue dynamics, its repeated accelerations and decelerations still mimic key soccer-specific demands [52]. Overall, we believe these tests provide ecologically valid simulations of match-induced fatigue and muscle damage.

While the included studies used a range of cold-water immersion protocols (8°C–15°C for 5–15 min), the majority aligned with established guidelines recommending 10°C–15°C for 10–15 min to optimize intramuscular cooling and user tolerance [53–55]. Moreover, evidence suggests that similar cooling effects can be achieved when thermal load (time × temperature) is equivalent, with a 1:1 ratio (e.g., 10 min at 10°C) commonly recommended [56]. Recent findings further indicate that variations in duration within this range may not significantly affect recovery outcomes [57]. Therefore, we believe that the protocol variability observed across studies reflects real-world practice and does not compromise the overall findings of this meta-analysis.

When taking the 95% PI into account, which encompassed the null value for all parameters, the overall findings should be interpreted with caution. Although the pooled effects on average favored CWI, the inclusion of future studies may attenuate or even nullify the observed benefits, reflecting variability across study contexts and populations.

Conclusions regarding the effect of CWI on DOMS at 72 h are based on a single study and should therefore be interpreted with caution. This, along with findings showing that several variables were still unrecovered at 48 h, suggests that future studies should incorporate longer follow-up periods for a complete recovery profile. Lastly, none of the included studies focused exclusively on female athletes, meaning that our findings may better represent male responses to CWI and highlight the need for more research on female soccer players.

5 | Conclusions

This systematic review with meta-analyses included ten original studies investigating the effect of CWI on recovery of physical performance, biochemical and perceptual outcomes following match or simulated match play in soccer players. Our findings indicate several potential benefits of CWI as a recovery intervention, including faster recovery of muscle strength and CMJ performance, accelerated normalization of CK levels, and reduced DOMS. Overall, CWI demonstrated greater effectiveness compared to passive recovery across all assessed time points (up to 72 h post-exercise). Given the high physical load and injury risks in competitive soccer, there is a strong need for evidence-based strategies to manage fatigue and muscle damage. Therefore, our novel findings provide valuable insights for practitioners and support staff, highlighting the potential of CWI in enhancing post-match recovery for soccer players. Nevertheless, given that the 95% PIs encompassed the null effect, the consistency of these benefits across future studies and settings remains uncertain.

6 | Perspectives

Our findings highlight the practical relevance of CWI as a recovery tool for soccer players, particularly in managing fatigue, muscle damage, and soreness in the critical 72 h post-match period. While the observed potential benefits on muscle strength, CK levels, and DOMS support its widespread use, the lack of improvement in sprint performance suggests that CWI may be more effective for certain recovery markers. Future research should explore optimal CWI protocols, including variations in water temperature, immersion duration, frequency, and timing relative to match play, between game cessation and the next game or training session to maximize its effectiveness. Moreover, more studies involving female soccer players are needed to explore potential sex-specific responses to CWI and recovery. Additionally, long-term studies assessing the cumulative impact of repeated CWI sessions over a competitive season could provide insights regarding its role in injury prevention and sustained performance during congested periods with several weekly matches. Furthermore, integrating CWI into a broader recovery strategy alongside nutrition, sleep, and active recovery may offer a more complete approach to player well-being, ensuring that athletes can maintain peak performance throughout the season.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** PEDro scores of included articles. **Data S1:** Supporting Information.